

CDS 6324
DATA VISUALIZATION

PROJECT (40%)

Lecture Section: TC2L

Tutorial Section: TT5L

Group Number: TT5L_G4

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1.0 Data Sources

The dashboard utilizes **WHO/UNICEF Joint Monitoring Programme (JMP) 2025 - Household Data**, an official United Nations database that tracks the progress on SDG 6 (Clean Water and Sanitation) worldwide. The dataset is obtained from washdata.org and contains data on 232 countries and areas between 2000 and 2024, with one entry per country per year.

The dataset includes four main elements such as Water, Sanitation, Hygiene, and Menstrual Health all of them having an identical format. The present project utilizes the **Water** and **Sanitation** datasets only as these two have the most direct connection to SDG 6 goals (access to safe water and sanitation for everyone); Hygiene and Menstrual Health were out of scope for our three analytical questions.

The database fully satisfies the minimum requirements of the project – 5,661 entries in the Water part and 5,651 in the Sanitation part, more than 50 attributes in both parts, as well as a temporal (year) and spatial dimensions (country and its SDG region and World Bank income group).

Scope decision: JMP produces a separate dataset (REG), containing aggregated regional information on top of the household-level world dataset (WLD), used in this project. We have used only the WLD file because it already contained SDG region and World Bank income group for each country, and thus did not need the REG file for regional and income-related analysis (Q1 and Q3).

Data cleaning: the raw JMP export included many columns irrelevant for answering our analytical questions (rural/urban breakdowns, different regional classifications schemes, and translations of country names into five languages). We removed all unnecessary columns in both CSV files, so that there were just ~12-19 columns left (e.g. country name, year, population, urban percentage, ISO code, SDG region, income group, and the safely managed). This way we decreased the overall size of the dataset from around 6 MB to about 1 MB, keeping our final submission below 2 MB mark without affecting any useful values.

2.0 Analytical Questions

The dashboard is built around **three** analytical questions, each answered by two complementary visualizations:

Q1. Which regions and countries are the most critically underserved in water and sanitation access, and what does that look like geographically?

- ➢ A spatial question answered by a choropleth map (the geographic pattern) paired with a ranked bar chart (the specific countries behind that pattern).

Q2. How has global access to clean water and sanitation progressed between 2000–2024, and is the world on track to meet the SDG 6 target by 2030?

- ➢ A temporal, forward-looking question, answered by an animated multi-line chart with a 2030 projection and target line, paired with a stacked area chart showing how the underlying service-level composition has shifted over time.

Q3. How does a country's income level relate to its water and sanitation coverage, and do wealthier nations consistently outperform poorer ones?

- ➢ A socioeconomic question, answered by a scatter plot (country-level spread) paired with a grouped bar chart (the average pattern by income group).

3.0 Visualisation

Viz 1: Choropleth Map (Q1)

Countries colored based on the degree of access to their safely managed water and sanitation facilities, darker being better. The user is free to zoom and pan for closer inspection of clusters, while the shading instantly changes upon moving the year slider or switching topics.

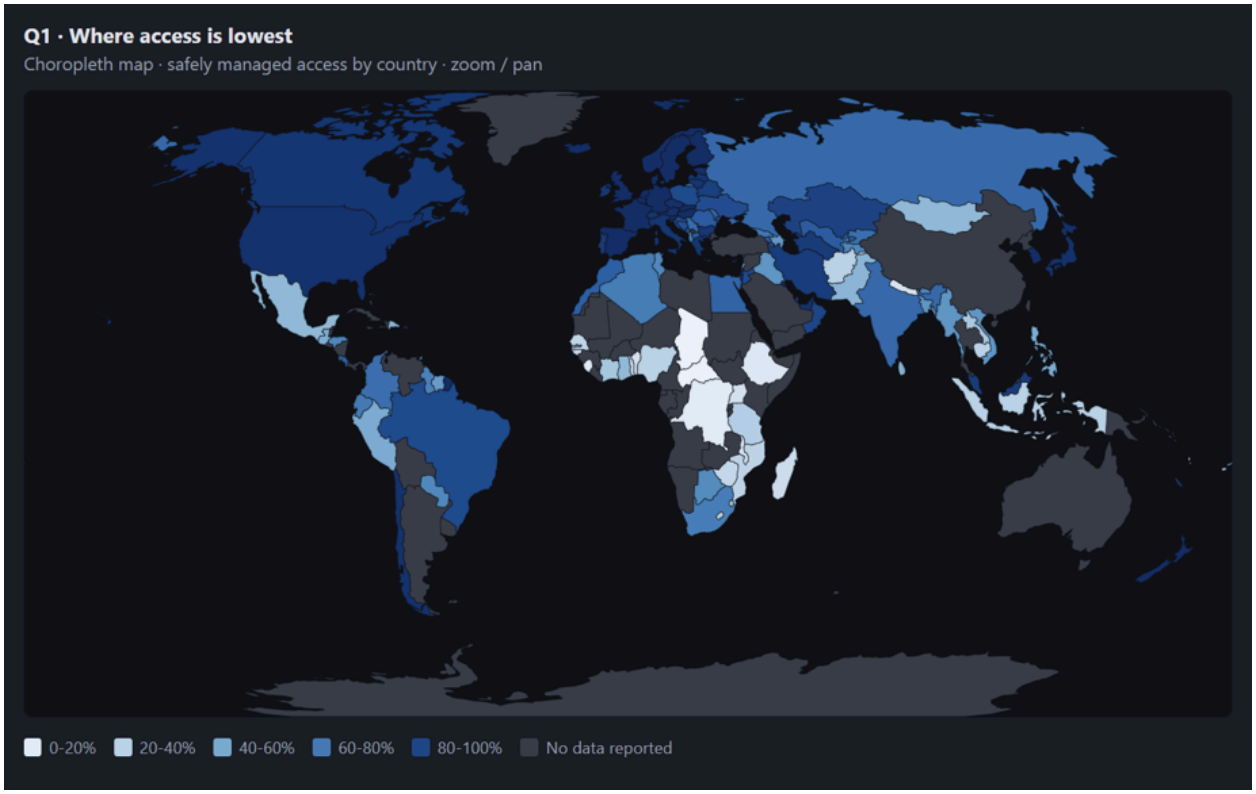


Figure 1 : Choropleth Map (Q1)

Key Insights :

- Lowest access countries are not randomly scattered on the map but form only one distinct cluster across Central and West Africa, making that the most noticeable problematic area on the map.
- Certain countries appear grey instead of being shaded, indicating that there is no access percentage data available for the particular year from JMP. It's worth calling out so those gaps aren't mistaken for 0% access.

Viz 2: Ranked Bar Chart (Q1)

It includes the list of the 10 countries with the safest managed access along with the 10 with the lowest access in order that the pattern shown on the map is connected to specific countries rather than regions represented with shades.

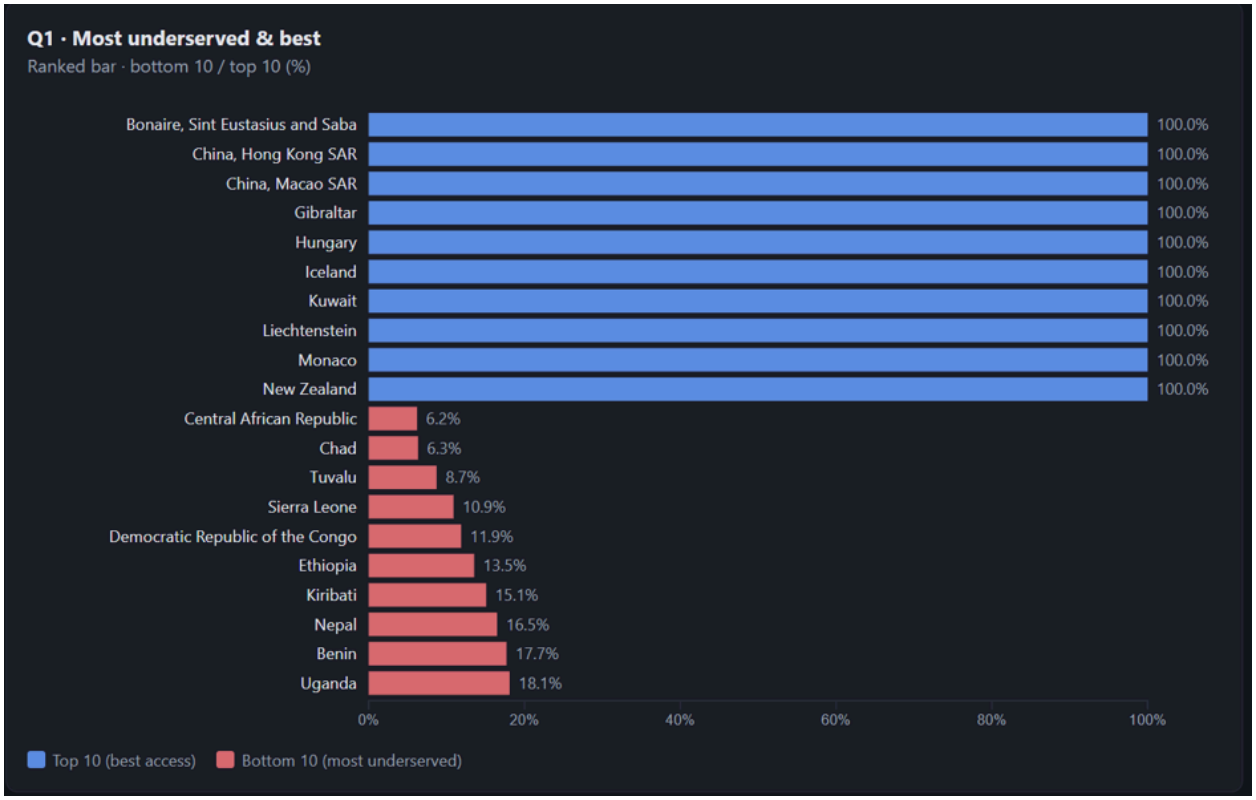


Figure 2 : Ranked Bar Chart (Q1)

Key Insights:

- The 10 leading nations are all located at 100% while the last nation on the list, Central African Republic, is only located at 6.2%, which is nearly 94 percentage points between the safest served nations and the least served nations.
- Eight out of the 10 countries with the lowest access are located in Sub-Saharan Africa, apart from the two outliers of Tuvalu and Kiribati, demonstrating that the cluster of underdeveloped countries seen on the map are the same few countries, not a widespread issue.

Viz 3: Multi-Line Chart (Q2)

This graph plots access to safe water and sanitation since 2000 to 2024, including dashed projections to 2030 and a reference line at 100% target. Pressing the Play button will animate the graph by year, which will also drive the year slider used throughout the dashboard.



Figure 3 : Multi-Line Chart (Q2)

Key insight:

- The access to water has always been greater than access to sanitation throughout the entire period, but the difference between them has been greatly reduced from approximately 20 percentage points difference in 2000 to only 4 in 2024.
- Despite being projected further in time, both lines land around at approximately 70% in 2030, far below the 100% target, which means that the current trend will not help to reach SDG 6 on time.

Viz 4: Stacked Area Chart (Q2)

Provides information about how the global population is distributed among the five JMP service levels (safely managed, basic, limited, unimproved, and no service) with regards to the chosen topic between the years 2000 to 2024 in order to understand the composition behind the Viz 3 trendline band by band.

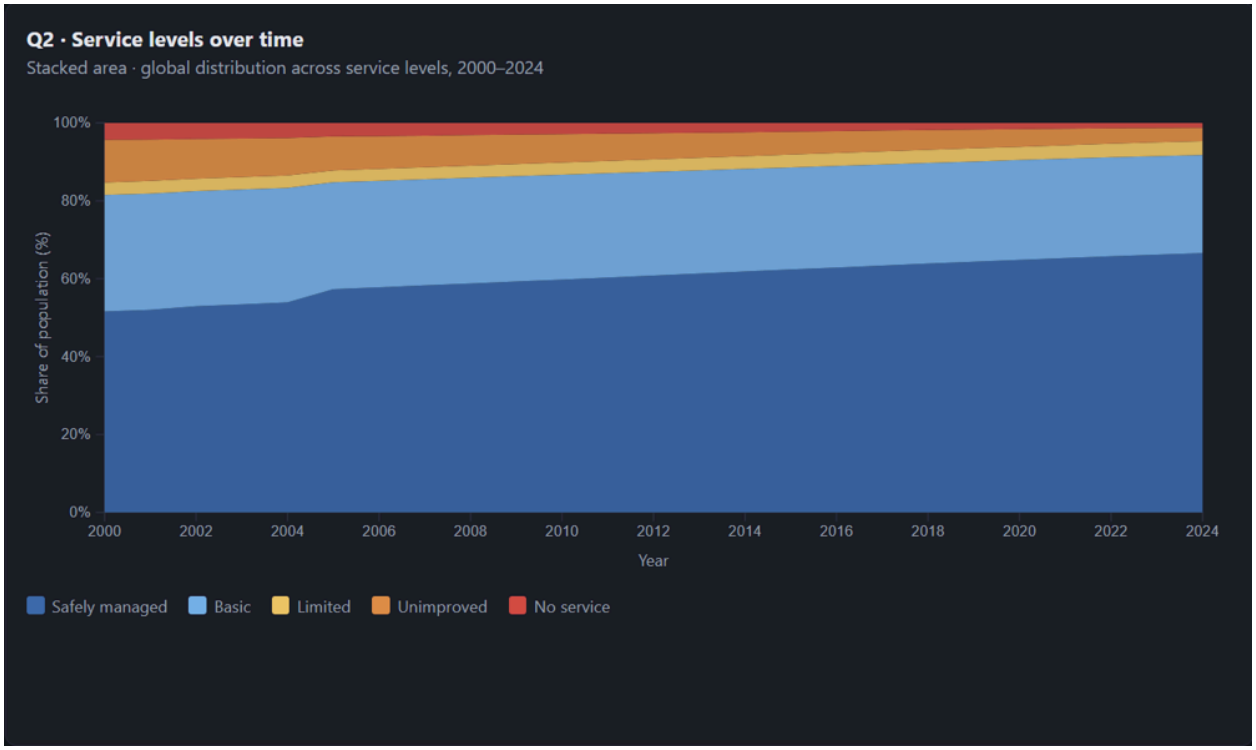


Figure 4 : Stacked Area Chart (Q2)

Key insight:

- The band under safely-managed increased consistently from approximately 52% to 66% of the global population and most of the increase was achieved through shrinking in the basic band – people who have climbed a step on the service ladder but not those who started from scratch.
- The three lower bands (limited, unimproved, no service) decreased marginally, from about 18% to 10%, indicating that even after 24 years of progress, one out of ten people is still without access.

Viz 5: Scatter Plot (Q3)

Each of the countries is represented as a dot in the plot – safely managed water access are plotted on the x-axis while safely managed sanitation access are plotted on the y-axis, population size as the size of the bubble and income category as the color, with a diagonal line marking equal water and sanitation access.

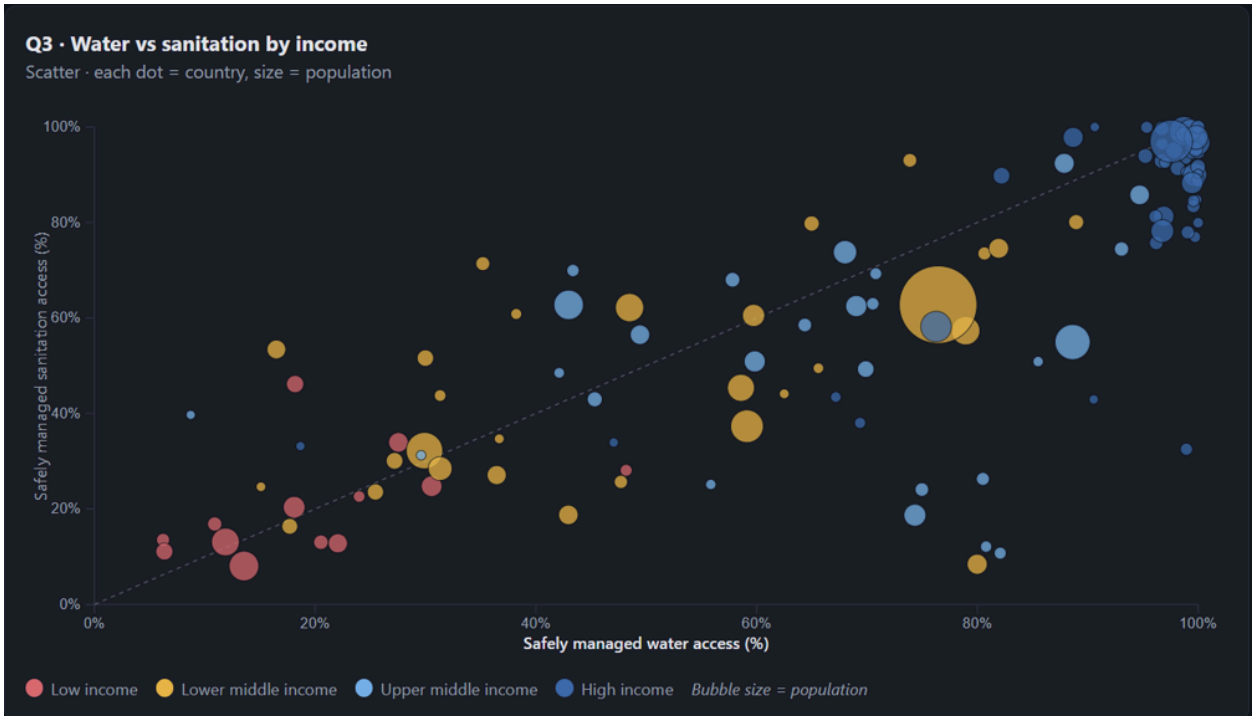


Figure 5 : Scatter Plot (Q3)

Key insight:

- The majority of the points can be found close to the diagonal line, which means that the country's water access level is a fairly accurate measure of sanitation access level since they do not change independently.
- The dots that lie farthest from the line (sanitation lagging water) consist of middle-sized bubbles and not largest ones, which means that the discrepancy is not restricted to most populous countries.

Viz 6: Grouped Bar Chart (Q3)

Measures the average proportion of safely managed water and sanitation facilities within each of the four World Bank income groups for the chosen year, quantifying how the extent of coverage changes with income group.

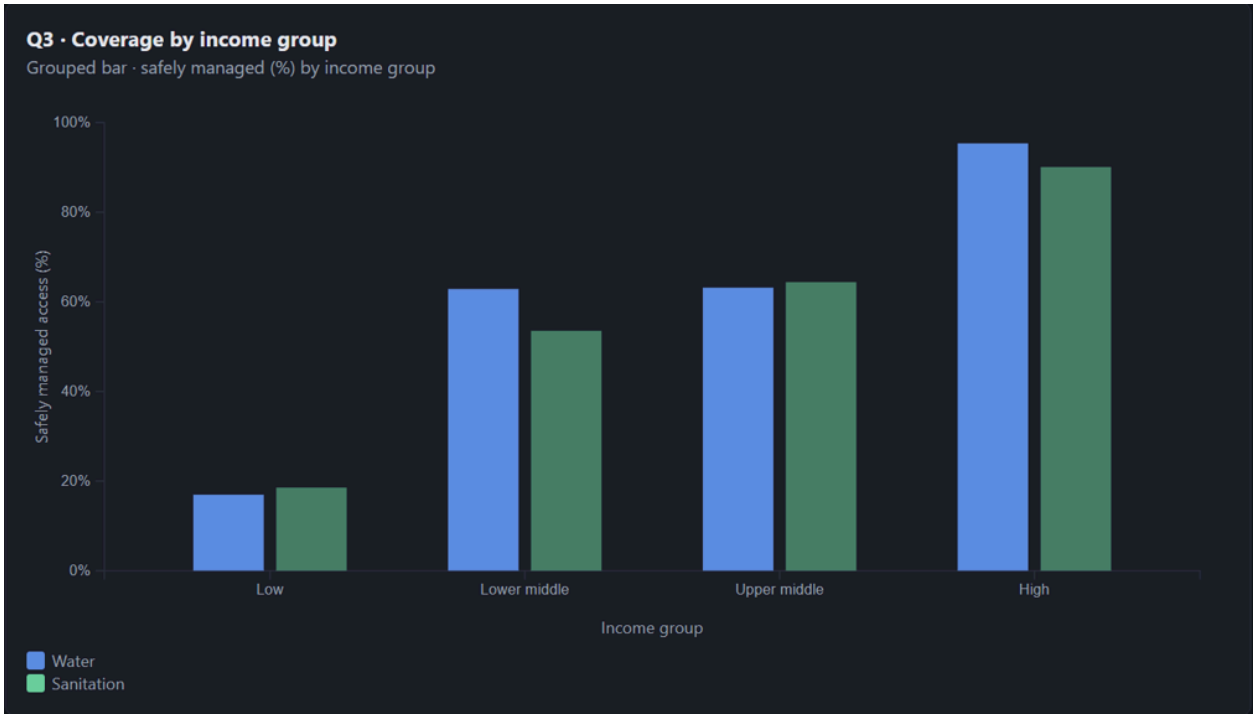


Figure 6 : Grouped Bar Chart (Q3)

Key insight:

- There is a direct relationship between the level of income and the increase in coverage, ranging from just 15% in low-income countries to more than 90% in high-income countries, where there exists a six-fold gap between the two extremes.
- In the upper middle-income category, sanitation levels are higher than those of water, thus being the only category in which the trend of higher water than sanitation does not hold.

4.0 Interactivity

The dashboard uses a single control bar (topic toggle, year range, region selector) located on the top that controls all of the charts and not individual control bars for each chart.

Hover tooltips

➢ Every visualization offers the user information through hovering, the map and ranking bar offer the country name, access percentage(%) and region; the line

and area charts offer the year and values for both of the topics; whereas the scatter and grouped bar visualizations offer the country/income group along with the water & sanitation information.

Filtering

- Topic toggle (Water/Sanitation) helps choose the correct charts according to the topic selected, while the region selector ensures that the ranked bar, stacked area, and grouped bar visualizations are displayed for only one region from the particular SDG.

Zooming and panning

- There is a zoom-in/pan feature on the choropleth map visualization that enables users to focus on specific country clusters (like Central Africa).

Linked interactions

- The slider for the year acts as the key element that, on being moved, refreshes the map, ranked bar graph, scatter plot graph, and the year line of the multiple line graphs in order to make sure that all the graphs represent the same period of time.

Animation

- The multi-series chart is such since the play button on this chart will automatically move from the year 2000 till the year 2024 moving the year slider forward at each step thereby creating animation in the map and the ranked bar graph using real data-driven movement rather than a decorative effect.

5.0 Best Data Visualization Practices

The dashboard follows Shaffer's 4Cs to guide the design decisions:

- **Clear:** The purpose of each individual graph is clear from its name and subtitle (e.g. "Q1 · Where access is lowest"). Each title and subtitle of a graph tells you what it shows.
- **Clean:** Consistent dark background, limited color usage (water – blue, sanitation – green, single accent color scale for income/rank) and absence of any unnecessary decorations like grid lines help focus on the information shown in the graph.

- ➢ **Concise:** Only the data that is necessary for answering an analytical question is shown in each individual graph, the ranked bar graph shows only the top/bottom 10 countries instead of all 232, and the multiple line graph only shows two lines for the topic and one line for the target.
- ➢ **Captivating:** Interactivity (hover tooltips, the animated year slider, zoom/pan on the map) makes graphs interesting to use and invites exploration instead of showing a static image.

6.0 Challenges & Solutions

- ➢ Global or regional aggregations not included in the data set. The JMP data set only includes a single row per year per country; there is no row for "World" that could be used to draw the graph directly. Solution: we computed the population-weighted average of all countries for each year ourselves, thereby making sure that large countries have more weight than small ones when computing the global trend.

- ➢ Matching country names in the data set with the map. The map relies on its own world boundaries data (in topojson format), which contains IDs for countries that don't match those used in the JMP data set. Solution: we created our own mapping of the map country IDs to their ISO3 codes and used that as the key field to join the data sets rather than relying on country names which might not always match (e.g. "Czechia" vs "Czech Republic").

- ➢ Maintaining synchronization of six graphs. Given that all graphs depend on the same set of filters (Topic, Year, Region) and would update independently otherwise, a desynchronization may occur easily. Solution: we centralised the filter status into a shared object that is subscribed to by all six charts, so a single change (e.g. dragging the year slider) updates all charts at once.

7.0 Task Distribution

Student Name	Analytical Question	Visualization	Contribution Percentage	Signature
SIVANANTHAN A/L SELIYAN	Q1	Viz 1 + Viz 2	33.33%	
LAXMAN PILLAI A/L RAMESH PILLAI	Q2	Viz 3 + Viz 4	33.33%	
KARRTHIK A/L NADARAJAN	Q3	Viz 5 + Viz 6	33.33%	

Laxman also integrated the six charts into a single dashboard, connecting them through the shared filters and layout so they work together as a unified system.

Collaboration Approach

At the initial meeting, the group decided on the color scheme and the design of the dashboard so that all the future charts will be consistent with it. Each of us then created his own two charts independently of others. When all the charts were completed, we had another meeting, where everyone presented his work and discussed how it was created. Then Laxman put all six charts into one nice dashboard. The report was also prepared collectively by all three of us, with each author responsible for his part corresponding to the charts. Overall the process went smoothly, as we were able to meet regularly and sort out any issues as they came up.

Time Spent

In total for the group, about 82 hours, with each member contributing for about 3 hours daily during the 4 weeks of the project.

Key Challenges

Discussed under the Challenges and Solutions section above, especially getting six graphs to work together in one filter and connecting the data set to the map countries.